

Section 1.3:

Operating Systems

This section will take a look at operating systems and their role in the computer system.

Operating systems are pieces of software that manage everything that happens in your computer and they instruct the hardware on what to do.

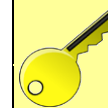
The operating system makes your system useful. Without it your computer would sit there and do nothing.

Learning objectives of this section:

- | | |
|---|---|
| # | Know the definition of an operating system and why computer systems need them. |
| # | Understand the different tasks that an operating system performs. |
| # | Be able to describe the different types of operating systems and be able to compare them to each other. |

What is an Operating System?

Definition: ***"An operating system is a software program that manages computer resources. Operating systems allow components to communicate with each other and enables the computer to run software applications."***



Key Words:

Operating System, Manage, Communicate, Resources, Inputs, Outputs.

Tasks of the Operating System

Operating systems perform the following functions:

- 1. Accepts inputs** from the mouse or keyboard.
- 2. Sends outputs** to the **monitor or printer**.
- 3. Recognizes peripheral devices** such as external hard disks, pen drive, web cam etc and makes sure that **software** needed for the hardware to run **is installed**.
- 4. Manages files and folders in the system** (Naming, Creating, Moving, Finding and Deleting folders etc).

5. Allows **applications software** (word-processing, spreadsheets etc) to **communicate with the system's hardware**.

7. **Loads** and **runs** software **applications**.

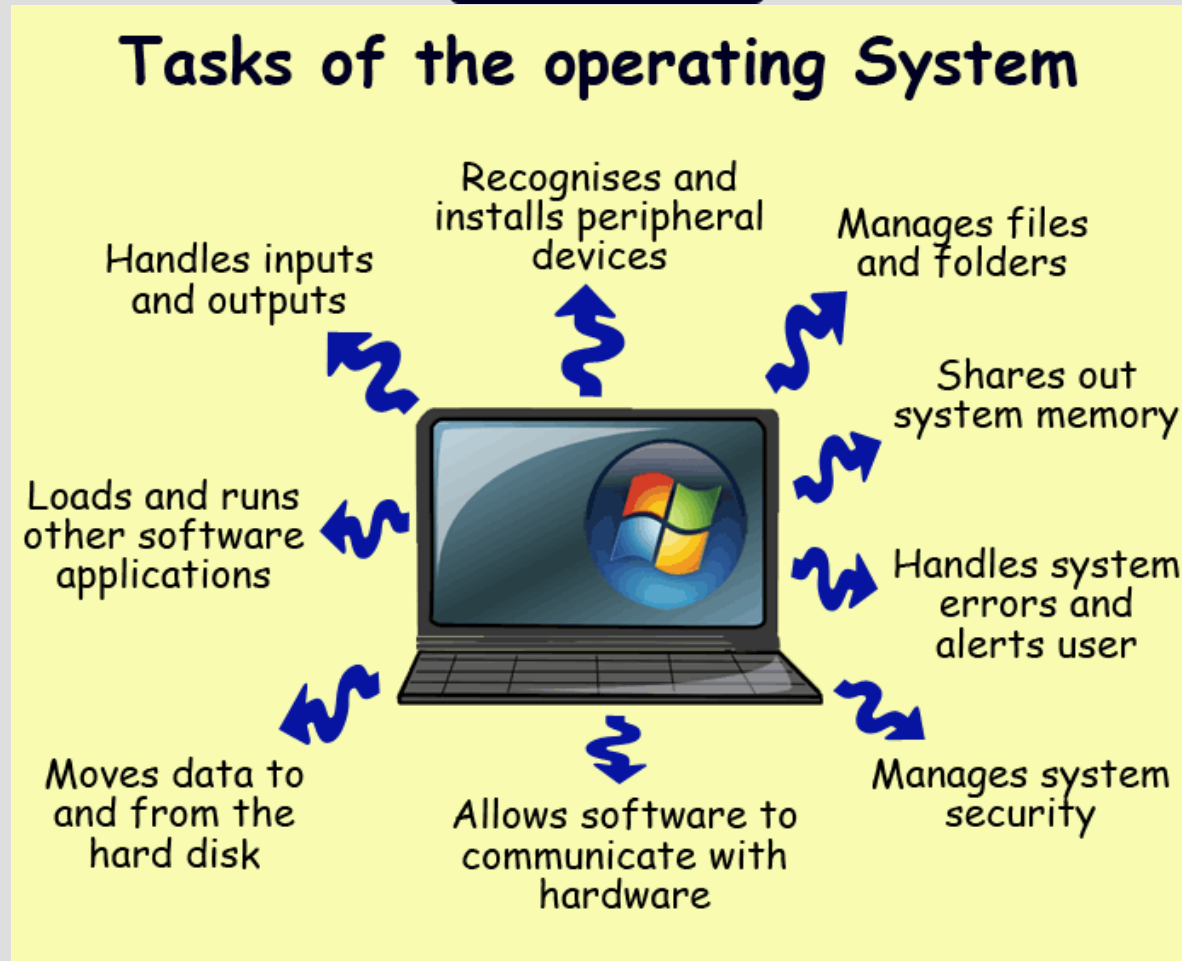
9. **Handles system problems and alerts the user**. For example if a printer is jammed and cannot printer, the operating system will stop the print job and alert the user with a warning message.

6. **Shares out system memory efficiently**. The operating system will decide how much memory to assign to particular tasks. It also moves data in and out of memory.

8. **Manages system security**. For example - allows passwords to be added / changed.

10. **Manages the moving of data** to and from a hard disk.

[Click for Picture Example](#)

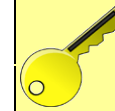


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Types of Operating System Interfaces

The way in which users communicate with the computer is called an '**interface**'. The interface is what we use to give the computer commands. There are **three types** of operating system interfaces:

- **Command Line Interface** (CLI)
- **Graphical User Interface** (GUI)
- **Touchscreen Interface**



Key Words:

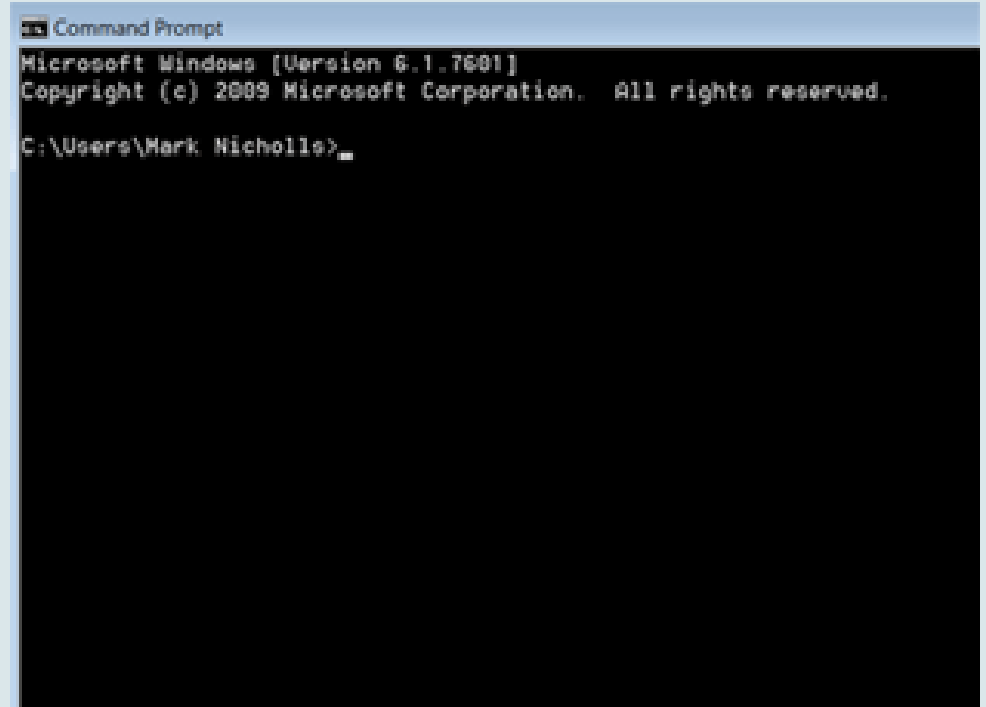
Graphical User Interface (GUI), Command Line Interface (CLI), Command Prompt, Windows, Icons, Menus, Pointers, Post WIMP.

1 Command Line Interface (CLI)

General Information:

- # A command line interface is an older style operating system where users type in commands using keyboard.
- # Command Line Interfaces **do not make use of images, icons or graphics**. All the user is sees is a plain black screen like the one to the right. →
- # Because they use no graphics they require **very little computer power**.
- # There are over **270 different commands** that can be entered at the command prompt. Commands have to be **entered precisely** without spelling mistakes or else the operating system will return an error.
- # Remembering commands and the exact way to enter them can be difficult and so Command Line Interface Operating Systems are **considered hard to use**.

Examples:

A screenshot of a Windows Command Prompt window. The title bar says 'Command Prompt'. The text inside the window reads: 'Microsoft Windows [Version 6.1.7601] Copyright (c) 2009 Microsoft Corporation. All rights reserved. C:\Users\Mark Nicholls>'. The background is black and the text is white.

```
Command Prompt
C:\Users\Mark Nicholls\Documents\My Files>dir
Volume in drive C has no label.
Volume Serial Number is 8CBE-2A36

Directory of C:\Users\Mark Nicholls\Documents\My Files

14/09/2012  16:10    <DIR>          .
14/09/2012  16:10    <DIR>          ..
14/09/2012  16:07                0 JUNK.txt
04/09/2012  19:26                75 osoul account.txt
14/03/2012  08:48            10,560 Southern Gothic Horrors.docx
24/02/2012  18:07            10,607 website domain and hosting.docx
11/12/2011  19:43                75 WIN 7 Pro.txt
               5 File(s)              21,317 bytes
               2 Dir(s)  35,198,140,416 bytes free

C:\Users\Mark Nicholls\Documents\My Files>del junk.txt_
```

Command Line Interface:
Commands are entered at the prompt
`C:\Users\Mark Nicholls>`

Command Line Interface

A Command Line Interface operating system works via the user entering **typed commands** with a keyboard. There is no use for a mouse.

Command Line Interfaces do not use windows, icons, menus or pointers.

There are over 270 commands available for functions such as delete, open, run etc.

Commands must be entered precisely with no spelling mistakes. These commands can be difficult to remember and, as a result, this type of interface is considered **more difficult to use** when compared to a GUI.

In the example on the left I am using a version of MSDOS to delete a file named **JUNK.txt**

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Example of a command line interface.
(click image to zoom to retrieve more information)

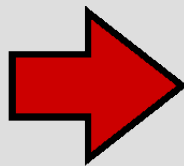
Key features of a Command Line Interface:

- # The main features of a CLI are that keyboards are used to type in a variety of different commands into a command prompt.

The sequence of images below will show this concept better:



Keyboard used to.....

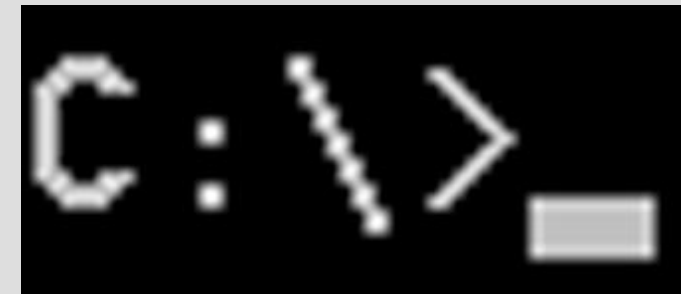
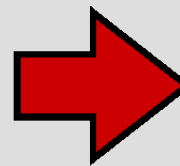


```
C:\Powershell>cd Glossary
C:\Powershell>dir
Volume in drive C has no label.
Volume Serial Number is FACC-E7D4

Directory of C:\Powershell

09/29/11  02:55 PM    <DIR>          .
09/29/11  02:55 PM    <DIR>          ..
09/29/11  02:52 PM    <DIR>          All
09/29/11  02:52 PM    <DIR>          Blog
09/29/11  02:52 PM    <DIR>          Facebook
09/29/11  02:55 PM    <DIR>          Glossary
09/29/11  02:52 PM    <DIR>          Twitter
09/29/11  02:52 PM    <DIR>          Website
               0 File(s)                0 bytes
               8 Dir(s)  64,644,661,248 bytes free
```

Enter commands into.....



A command prompt

Examples of some commands:

- # There are **many commands** that you can enter into a command line interface.

In fact, if you can think of something you want the operating system to do, there is a good chance that there is a command for it!

- # The table to the right shows you some **common CLI commands** and also describes what they do.



Activity!

Do some research and **find 5 different commands** that have not been mentioned on this page.

Type the commands up, making sure to include the name of each command and a short description of what they do.

Command	What it does
copy	Copies files from to another location
del	deletes one or more files
format	Deletes all the data on a hard disk
md	Creates a new folder
rename	Renames a file or folder

Remember: there are over 270 of these commands.

2 Graphical User Interface (GUI)

General Information:

- # GUI's are visual (graphical) interfaces and they are more popular than CLI's because they are **very easy to use**. The graphics do need more computer power however.
- # Instead of typing in commands, the user can use a mouse to **point and click objects on the screen**.

Examples:

For example: A user can **erase a file** by **right clicking** and then **selecting delete**.

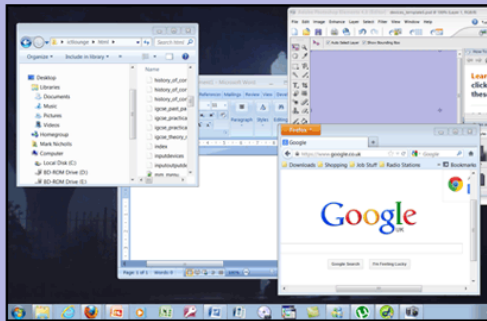
Key features of a Graphical User Interface (GUI):

The main features of a GUI are **Windows, Icons, Menus** and **Pointers**.

This is often abbreviated to **WIMP**.

The table below describes these 4 features in more detail:

Windows



Windows

Windows are one of the features of a GUI operating system.

Windows divide the screen into areas so you can work on several tasks at the same time.

Windows can be made up of folders, applications, videos, web pages etc.

In the example on the left you can see a variety of different windows open at the same time.

The user can divide the screen into **separate areas known as 'windows'**. These windows can be anything from folders to software applications.

Windows allow you to work on several tasks at the same time.

Icons

[Click for more info](#)



Graphical User Interface

The graphical user interface (or GUI) is a type of operating system that makes use of:

- # Windows
- # Icons
- # Menus
- # Pointers.

Icons and options on menus represent folders, applications and other commands which are activated when selected and clicked by the pointer.

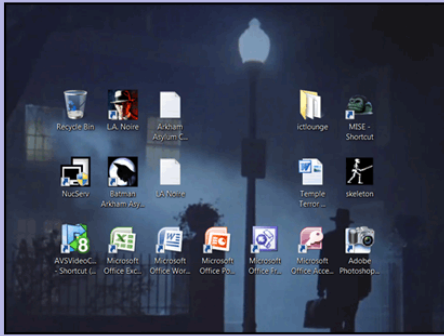
A mouse is normally used to direct the pointer around the screen.



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Example of a Graphical User Interface.
(Click image to zoom)

Key features of a GUI:



Icons

Icons are **small images** or **symbols** that represent **files, commands or windows**.

Icons can be **clicked to activate a command** or **open a folder** etc.

In the example on the left you can see a variety of different icons in different positions on a desktop.

Some of the icons represent folders and others are shortcuts to applications and games.

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Icons are '**symbols**' or **small images / graphics** that are used to represent files and commands. Icons can be clicked to carry out a function or open a file etc. The user operates a mouse to move a pointer over the icon and then clicks it to activate the function.

Icons are also used to **represent folders**.

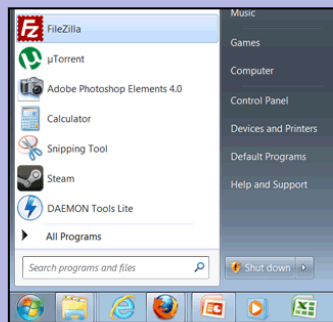


Mouse used to control and command a pointer



Icons can be clicked to open folders or activate commands

Menus



Menus

Menus allow users to select commands and icons from a drop-down list.

Menus are normally either **pop-up** or **drop-down** in style and usually disappear when not in use. This saves space on your desktop.

In the example on the left you can see some icons that are part of a menu.

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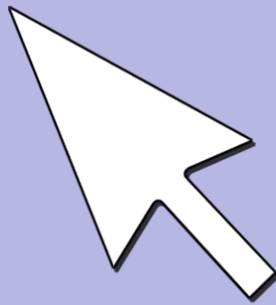
These allow users to **select functions from a list**. Each item in the list will perform a different function.

Menus usually either '**pop-up**' or '**drop-down**'. The navigation used at the top of this web site are examples of drop-down menus.



Menu items can be selected using the pointer.

Pointers



[Click for more info](#)

Pointers

Pointers usually look like arrows and are used to select icons and the options found in menus.

The pointer is moved around the screen via a mouse. The mouse buttons are used to activate icons and menu options once the pointer is in place.

The shape of the pointer can sometimes change depending on the application you are using at the time.

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Pointers are little arrows that you move across the screen by directing your mouse. Pointers can be used to:

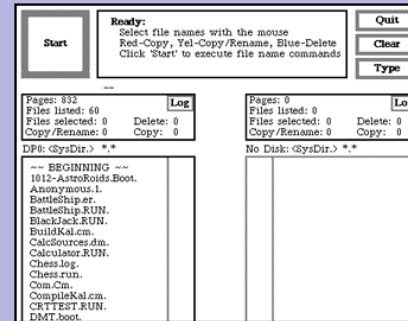
- **Select and use icons**
- **Select options found in menus**
- **Reposition folders and icons on the screen.**

NOTE!

GUI operating systems were first created by **Xerox** in **1973**. The idea was further developed by Apple on their Macintosh system.

[Click for Picture Example](#)

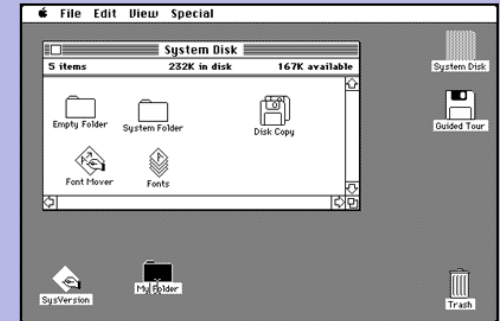
Xerox Alto GUI (1973)



The first modern GUI was developed at Xerox Palo Alto Research Centre (PARC) in the early 1970's.

The operating system was first found on the Xerox Alto which was used as a research computer in Universities.

Mac OS System 1.0 (1984)



System 1.0 was the first GUI style operating system developed for the Apple Macintosh.

System 1.0 was heavily influenced by the Xerox GUI and made use of, now familiar, mouse-driven WIMP features (Windows, Icons, Menus, Pointers).

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3 Touchscreen Interfaces (aka Post-WIMP Interface)

General Information:

- # Portable devices such as **mobile phones**, **PDA's** and **tablets** (e.g. iPad) use **interfaces similar to a GUI** (WIMP) where **icons and menus** are used to input commands.

Examples:

However, because these devices **don't have room for a mouse**, the way in which the icons and menus are used is different.

Touchscreen technology allows people to **use their fingers** to select **icons** and **options** straight from the device's screen.

We call this type of interface **Post-WIMP**.

Key features of a Touchscreen Interface:

Features available on POST-WIMP interfaces are highlighted in the table below:

Pinching

[Click for Video Example](#)

This is where you **pinch your fingers together** across a touchscreen to **zoom into** an image, application or document on your device.

You can also move your **fingers apart** to **zoom out**.



Post WIMP Interface

Post WIMP Interface's are found on operating systems used on mobile devices such as:

- # Mobile Phones
- # iPods
- # iPads
- # PDA's

These devices **do not have room for a mouse** and so they need different ways for the user to select icons and menus etc.

Touch screens are used so that the **user's finger** takes the place of the **mouse and pointer**. 'Fingers can be moved in different ways (called gestures) to interact with the operating system:

- # Zoom into images / documents / web pages
- # Rotate images
- # Select icons
- # etc

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Example of a Post WIMP Interface.
(Click image to zoom)

Rotating

[Click for Video Example](#)

This is where you use **two fingers** - one finger moves up while the other finger moves down the touch screen to **rotate an object**.

Swiping

[Click for Video Example](#)

Swiping is where you **swipe your finger** across the touchscreen to **scroll** through a document or turn the page.



Modern mobile phones often use touchscreen technology.

Comparing Command Line and GUI Operating Systems

Both command line and graphical user interface style operating systems have their **advantages** and **disadvantages**. The table below explains some of these:

Command Line Interface		Graphical User Interface	
Advantages	Disadvantages	Advantages	Disadvantages
User directly communicates with the computer.	Users must remember complex commands.	Quicker to enter commands. (You just click icons etc)	Smaller range of commands can be used (Icons are pre-programmed to set tasks and users cannot change this)
A wider range of commands can be used.	Lots of typing needed for quite simple tasks.	Easier to enter commands (You don't need to remember anything)	

Needs **very little** computer power.

(This operating system can be run on very old computers)

Higher **chance of errors** when typing in commands.

(One spelling mistake means that the command will fail)

Less chance of users making errors (Just clicking icons)

Graphical user interfaces require **more computer power**.

Revision
tips!

How to remember the difference between Command Line and GUI operating systems.

1. Can you use a mouse to control a pointer? **YES**
2. Can you select and click icons and menu options? **YES**

Then you are using a **GUI**

1. Can you use a keyboard to enter typed commands? **YES**
2. Can you use a mouse to control a pointer? **NO**
3. Can you select and click icons and menu options? **NO**

Then you are using a **COMMAND LINE INTERFACE**

Activities!

[Operating Systems - Task 1](#)

Click the above task and fill it in to show your understanding of the two main types of operating systems.

[Operating Systems - Task 2](#)

Fill in the above sheet using the links provided.

Examples of Command line Interfaces and GUI's

Command Line Interfaces

Unix

```
*** Welcome to Dante ***
Your computer for easy access to electronic mail and information.

Dante - electronic mail and information access
Press one of the following keys:

E - email: Electronic mail (Pine version 4.44)
U - internet: UW Homepage, via Lynx
H - home: About Dante, the Student Technology Fee, guidelines
A - announce: Announcements from UW Computing & Communications
O - other: Other choices
S - shell: To enter Unix commands
L - logout: End session

For use by authorized UW account holders only.
NOT FOR COMMERCIAL USE.
For policy on privacy and monitoring of accounts, type H (About Dante).
```

IBM DOS

```
Microsoft(R) Windows DOS
(C)Copyright Microsoft Corp 1990-2001.

C:\>mem

655360 bytes total conventional memory
655360 bytes available to MS-DOS
578352 largest executable program size

4194304 bytes total EMS memory
4194304 bytes free EMS memory

19922944 bytes total contiguous extended memory
0 bytes available contiguous extended memory
15580160 bytes available XMS memory
MS-DOS resident in High Memory Area

C:\>
```

Graphical User Interfaces

Windows 7



Windows XP



Mac OS X



Linux